UFSM00066 · Sistemas Dinâmicos
Lista de exercícios 6
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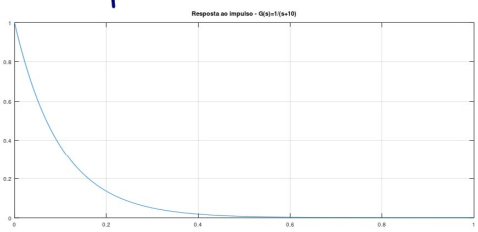
Sistemas de primeira ordem

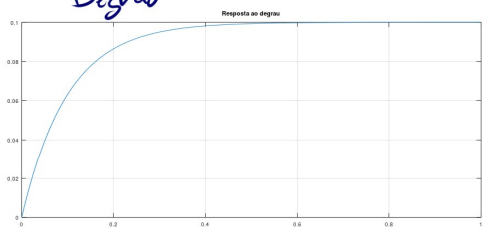
1. Desenhe aproximadamente a resposta ao impulso, ao degrau e a rampa do sistema cuja FT é a seguinte:

$$G(s) = \frac{1}{s+10}$$

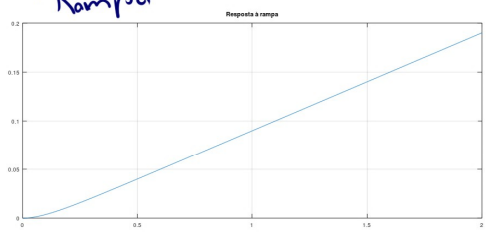
Impulso



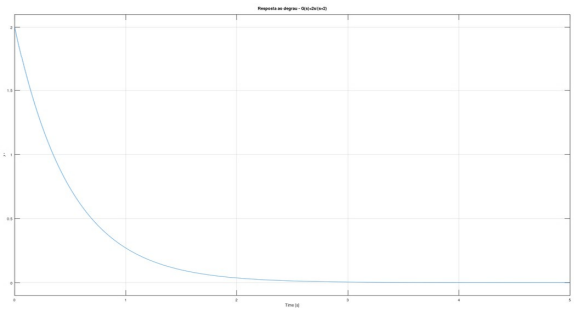
Degrau



Rampa

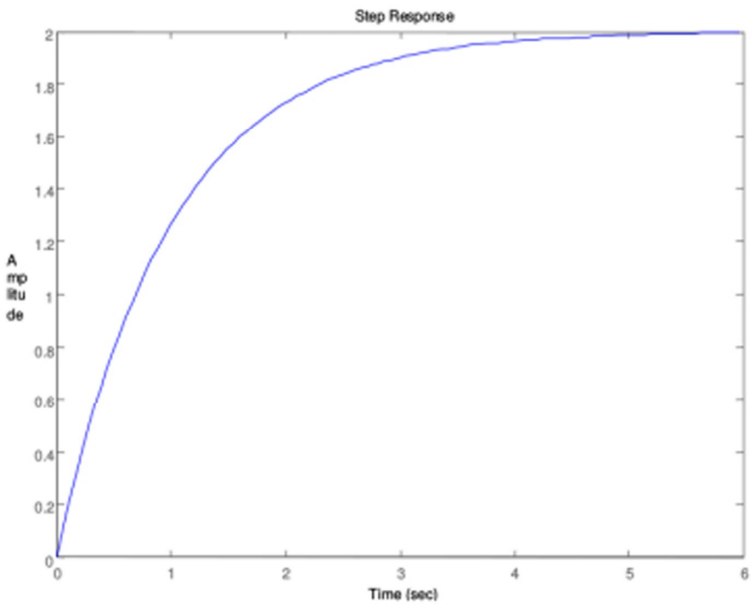


2. Desenhar a resposta ao degrau do seguinte sistema: $G(s) = \frac{2s}{s+2}$



3. A figura abaixo apresenta a resposta a uma entrada degrau, cuja função. Obter a resposta do sistema ante uma entrada impulso.

$K=2$



$\tau \approx 63{,}2\%$

$L \cdot 0{,}632 \approx \underline{1{,}26}$

$\tau = 1$

$F.t \rightarrow G(s) = \frac{K}{\tau s+1} = \boxed{\frac{2}{s+1}}$

$y(t) = 2e^{-t}u(t)$

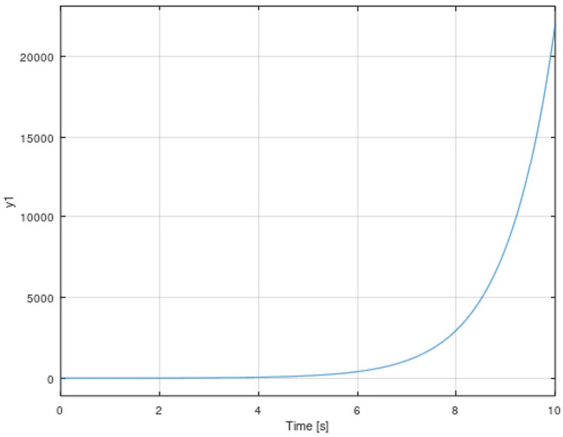
4. Dado o sistema $G(s) = \frac{5}{3s}$, realimente o sistema e obtenha a FT equivalente. Se este sistema trata-se da carga e descarga de uma caixa d'água, obtenha os valores de R e Q.

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{5/3s}{1+5/3s} = \frac{5}{3s+5} = \frac{1}{,6s+1}$$

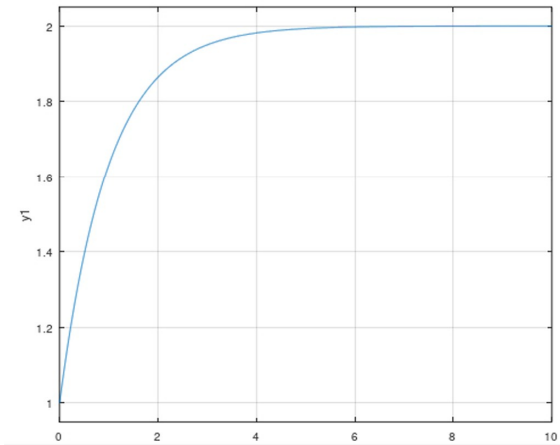
$$\frac{R}{RCs+1} \quad \Big| \quad RC = 0,6 \quad R = 1 \quad \rightarrow \quad \underline{\underline{C = 0,6}}$$

5. Desenhe a resposta ao degrau dos seguintes sistemas:

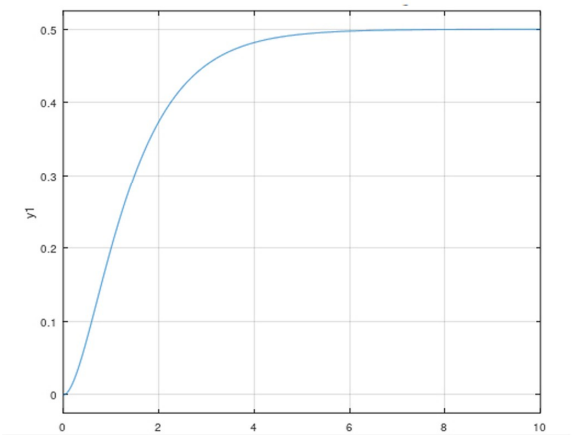
$$G_1(s) = \frac{1}{s-1}$$



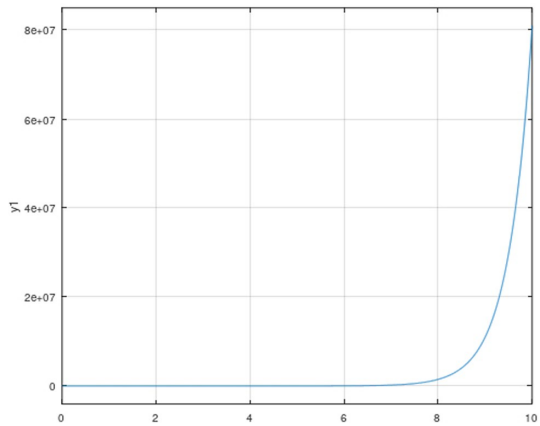
$$G_2(s) = \frac{s+2}{s+1}$$



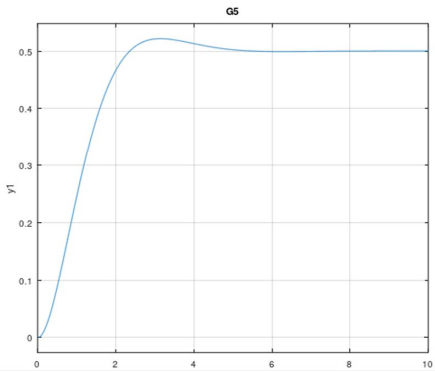
$$G_3(s) = \frac{1}{(s+1)(s+2)}$$



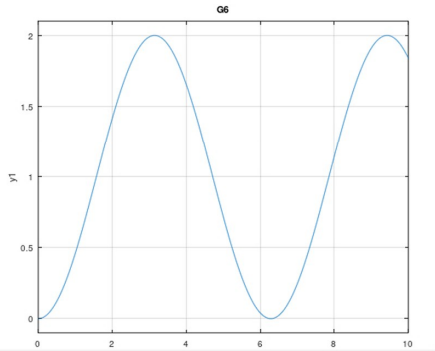
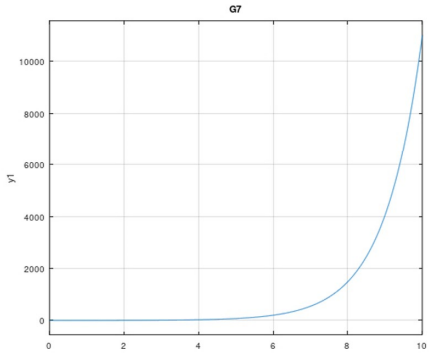
$$G_4(s) = \frac{1}{(s+1)(s-2)}$$



$$G_5(s) = \frac{1}{(s+1+j)(s+1-j)}$$



$$G_6(s) = \frac{1}{s^2+1}$$



$$G_7(s) = \frac{1}{s^2-1}$$